An Ensemble Classification Approach for Recognizing Steady-state Visually Evoked Potentials Frequencies

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***Abstract*—Brain-Computer Interface (BCI) is an interface that allows controlling computers using brain activity. BCI is capable of providing control commands that are hands-free and voice-free to increase the social inclusion of disabled people and to substitute typical devices. One common BCI paradigm is the Steady-state Visually Evoked Potentials (SSVEPs), which represent the response of the brain to flickering visual stimuli. This study proposes an ensemble classification algorithm to optimize the recognition of visual stimuli corresponding to the recorded SSVEPs. The algorithm considers the predictions of multiple classification and pre-processing approaches by taking their votes. Brain signals vary, not only from subject to subject, but also from one day to another for the same subject. Thus, most research work considers subject-dependent analysis. In this study, we evaluate the proposed approach both individually within each subject and across subjects. The algorithm is also tested for different stimulation times. The results demonstrate that for subject-dependent testing, the ensemble classification technique outperforms other approaches reaching a mean accuracy of 96.73% and 93.45% when using 5 sec and 2.5 sec stimulation times, respectively. For subject-independent testing, the ensemble classification technique achieved a mean accuracy of 92.18% and 76.91% when using 5 sec and 2.5 sec stimulation times, respectively. These results indicate the utility of the proposed approach in enhancing the performance of SSVEP-based BCIs.**

I. Introduction

Brain-Computer Interface (BCI) is a type of neural interface that aims to provide a link between the nervous system and the outside world through recording from the brain or through stimulating the brain [1]. Electroencephalography (EEG)-based BCI is a commonly used non-invasive technique [2, 3], and it provides a signal with a fine temporal resolution [4]. Now that technology is becoming more and more sophisticated, non-invasive techniques are the most widely used for both disabled and healthy subjects for a variety of applications.

Types of control signals used in BCI systems can be divided into three groups: evoked signals, spontaneous signals, and hybrid signals [5]. Evoked signals are generated unconsciously by the user when a visual stimulus is presented. Spontaneous signals are elicited voluntarily by the user without the need for external stimulation. Hybrid signals are a mixture of different signals to utilize the advantages of multiple types. The two main types of evoked signals are P300 and Visual Evoked Signal (VEP) [5]. P300 is a signal that is elicited in the brain 300ms after an oddball visual stimulus is presented among various visual stimuli. A typical P300 application is the P300 speller, where letters are displayed in a matrix form. When the row or column that has the desired letter flashes, a P300 signal is generated [6]. VEP signals, on the other hand, are generated by the brain when the subject perceives periodic flickering visual stimuli [7]. In response, a VEP signal is generated in the visual cortex, and frequencies present in the recorded brain signal will be consistent with the frequency of the stimulus. VEP can be further classified into transient VEPs (TVEPs) and steady-state VEPs (SSVEPs). TVEPs are generated when the frequency of the stimulus is below 2Hz, whereas SSVEPs occur when the frequencies are greater than 6Hz [8, 9].

Using SSVEP-based BCIs, users can select a target by gazing at it. The BCI system then identifies the target based on the features of recorded SSVEP. The commonly used graphical user interface for SSVEP-based BCI is to have flickering buttons with unique frequencies, and each button represents a certain command. When the subject focuses on one of the buttons, the brain produces a signal with a frequency equal to the flickering frequency of the button. There are various types of SSVEP pre-processing methods to extract SSVEP features, but there are mainly two types: frequency filtering and spatial filtering [10]. Frequency filtering filters the signal based on its frequency characteristics. Two types of frequency filters used for SSVEP are bandpass filter and notch filter [10]. On the other hand, spatial filtering creates a combination of multiple channels to augment the SSVEP responses. There are multiple types of spatial filtering including Maximum Contrast Combination (MCC) that maximizes the ratio between SSVEP and background signals [11], and Independent Component Analysis (ICA) which takes multi-channel EEG data and separates it into independent components [12]. In addition to filtering, a typical SSVEP processing pipeline would include Principal Component Analysis (PCA) to reduce the dimensionality of the recorded data [10]. Common Spatial Pattern (CSP) is another approach that has been used to represent the signal analytically by reflecting the amplitude and phase information of the signal [10]. For identifying the corresponding frequency of an SSVEP pattern, multiple techniques have been proposed including Canonical Correlation Analysis (CCA), which computes the

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relation between multivariate datasets after linear combinations of linear data, and machine learning classification techniques [13, 14].

In this paper, we propose an ensemble algorithm that combines 8 different techniques to classify SSVEP data. We investigate the performance of the proposed approach when applied to data representing 5 sec stimulation time as well as a shorter stimulation time of 2.5 sec. We examine the performance when the training and testing of the algorithm are performed using the data of each subject only (subject-dependent) as well as the more challenging case of training the algorithm using data of a group of subjects and testing it using the data of another subject (subject-independent). The achieved results demonstrate the efficacy of the proposed ensemble approach in comparison to using each of the 8 techniques individually.

II. Methods

A. Dataset

The dataset used in this study is the EEG SSVEP Dataset III created by the Multimedia Authoring and Management using your Eyes and Mind (MAMEM) project [15]. The dataset is recorded from 11 subjects who conducted an SSVEP-based BCI experiment with 5 frequencies as the visual stimuli. EEG signals were captured via a wireless connection using the Emotiv Epoc headset that records 14 channels using electrodes distributed on the scalp with positions AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4. The frequencies of the visual stimuli were 6.66Hz, 7.5Hz, 8.57Hz, 10Hz, and 12Hz. The experiment is divided into sessions: The first one has 12 trials and the second one has 13 trials separated by 30 seconds resting period. For each trial, the visual stimuli flicker for 5 seconds and stay still for 5 seconds so that the subject rest his/her eyes.

B. Pre-processing and Feature Extraction

The pre-processing done in this study involves first applying the common average reference (CAR) filter. The CAR filter is a spatial filter that removes the common values among the electrodes and leaves only the deviation on each electrode by subtracting the mean of the signals of all electrodes from each electrode [12]. This process improves the signal-to-noise ratio (SNR) of the signal. The CAR filter has been used in previous studies and showed efficacy in denoising EEG signals [12, 16, 17]. It can be defined as

$$M_i(t)^{CAR} = M_i(t)^{ER} - \frac{1}{n}\sum_{j=1}^{n} M_j(t)^{ER} \quad (1)$$

where n represents the number of electrodes, $M_i(t)^{ER}$ represents the potential difference between the i^{th} electrode and the ear-reference. For each electrode i, the mean is subtracted from the potential difference, resulting in $M_i(t)^{CAR}$.

After applying the CAR filter, the pre-processing proceeds with channels O1 and O2 only as they record brain activity from the visual cortex that is responsible for processing visual inputs. Both channels have been used in previous studies and shown to give better results than using all channels [18–20].

Next, Fast Fourier Transform (FFT) is applied to the filtered data. FFT calculates the Discrete Fourier Transform (DFT) from a sequence of points. Fourier Analysis is used in this process to convert signals from the time-domain to the corresponding frequency-domain, resulting in the power spectrum of the components of the signal at different frequencies [21]. FFT is a faster version of DFT. DFT is computed as

$$F_n = \frac{1}{N}\sum_{m=0}^{N-1} f_m e^{\frac{-i2\pi mn}{N}} \quad (2)$$

where N is the number of samples. This implies that for each N input values of f_m there are N output values, increasing the computational cost to N^2. On the other hand, FFT decreases the number of performed computations, which is based on divide and conquer method: Every signal is divided into two smaller signals, where DFT is applied to the smaller ones, and the results are added together [22]. By doing so, for every input, the number of operations is reduced by half, decreasing the computational cost to $Nlog_2N$.

From the resulting power spectrum, only the frequencies of the stimuli and their harmonics are extracted. As the peak is not necessarily visible at the stimulus flickering frequency, a window is taken around each frequency and its harmonics. This window spans 1Hz before the frequency and 1Hz after the frequency. Next, the power spectrum is normalized to avoid biasing the results to features of wider ranges. One way to normalize the data is using the Z-score [23]. The Z-score calculates how many standard deviations a sample is from the mean. The Z-score is used in this study, and it is defined as

$$z = \frac{x-\mu}{\sigma} \quad (3)$$

where x is the data point, μ is the mean, and σ is the standard deviation.

Finally, Principal Component Analysis (PCA) is applied to the result. PCA is an orthogonal linear transformation that reduces the data dimensions without losing information [24]. It transforms the data to a new coordinate system such that the greatest variance of the data lies in the first dimension through some scalar projection, and the second greatest variance lies in the second dimension, and so on. In this study, the first 50 principal components are extracted prior to applying the classifiers when PCA is used.

C. Classifiers

Two main classifiers are used in this study: Support Vector Machine (SVM) and Random Forests classifiers.

1) Support Vector Machine: SVM has been used in many studies to classify EEG data. Previous studies have shown that its performance is better than other techniques such as neural networks [25]. The SVM distinguishes between two classes by finding the best separation boundary between both classes. The optimal decision boundary is the one with the largest margin. Support vectors are those data points that the margin pushes up against [26]. Data that cannot be linearly separated are transformed into another space using kernels. For simplicity, we present the mathematical equations for the linear SVM classifier. The maximum margin can be formulated as follows

$$margin \equiv \max \frac{|\boldsymbol{x} \cdot w + b|}{\sqrt{\sum_{i=1}^{d} w_i^2}} \quad (4)$$

where $\boldsymbol{x}$ is the data point, w is the decision hyperplane normal vector, and w_i is the i^{th} component of w, b is the bias term and d is the number of dimensions of the data [27].

To correctly classify the data using SVM, the following constraints can apply for training set (x_j, y_j)

$$\begin{aligned} w^T x_j + b \geq 1 \; if \; y_j = +1 \\ w^T x_j + b \leq -1 \; if \; y_j = -1 \end{aligned} \quad (5)$$

where y_j is the target value of point x_j. In the SVM, the Gamma parameter determines the slope or width of a kernel function. The lower/higher the gamma, the lower/higher the curve will be. The C parameter is a regularization parameter that controls the tolerance of outliers. In our study, the linear and polynomial kernels showed better performance compared to the other kernels. Gamma is set to 0.01 and C is set 1.

2) Random Forest: Classifying EEG signals using Random Forest also showed promising results in previous studies [28, 29]. A Random Forest consists of building blocks of Decision Trees, referred to as *predictors* in the context of Random Forest. Each predictor classifies the data point of interest. The class with the maximum predictions becomes the output of the Random Forest classifier. The most frequently predicted class can be defined as

$$f(x) = \arg\max_{y \in \mathcal{Y}} \sum_{j=1}^{J} I(y = h_j(x)) \quad (6)$$

where J is the number of independent predictors and h_j is the j^{th} predictor, and $f(x)$ is the prediction function for predicting y. The prediction function is based on a loss function $L(y, f(x))$ that minimizes the expected loss value $E_{xy}(L(y, f(x)))$ for joint distribution x, y. One common choice of L is the squared loss

$$L(y, f(x)) = I(Y \neq f(x)) = \begin{cases} 0 & \text{if} \;\; y = f(x) \\ 1 & \text{otherwise.} \end{cases} \quad (7)$$

Random Forests is based on the binary recursive partitioning trees where the predictor space is split into a sequence of binary partitions [30].

The best possible split is chosen based on a splitting criterion. One of the splitting criteria for the Random Forest classifier is the Entropy criterion, which is calculated at each node within each predictor as

$$E = -\sum_{i=1}^{c} p_i \times log(p_i) \quad (8)$$

where c is the number of unique classes and p_i is the prior probability of each given class. Max-depth in our work was set to 6 and the criterion was set to entropy.

D. Ensemble Classification

The ensemble classification approach proposed in this study involves combinations of the pre-processing and classifiers mentioned: Four varieties of each classifier are implemented based on the pre-processing: using PCA and CAR filter, using CAR filter only, using PCA only, and without using PCA or CAR filter. This leads to eight variations: four for the SVM classifier and four for the Random Forest classifier. For each test input, each of the eight classifiers predicts an output, and the output with the maximum number of votes is the output of the ensemble classifier. In this study, we perform two forms of the ensemble classifier: the unweighted ensemble classifier and the weighted ensemble classifier. The unweighted ensemble classifier gives each classifier equal weights during the voting process. On the other hand, the weighted ensemble classifier gives more weight to classifiers with higher training accuracy and less weight to classifiers with less training accuracy. Thus, the output of the weighted ensemble classifier is defined as

$$Class = \arg\max_{i} \sum_{j \in C_i(x)} W_j \quad (9)$$

where $C_i(x)$ is the set of classifiers that predicted point x to be a member of class i and W_j is the training accuracy achieved by classifier j.

E. Performance Evaluation

The performance of the ensemble classifier is evaluated using a subject-dependent approach and a subject-independent approach. For the subject-dependent approach, the data of each subject is split into 60% training, 20% validation, and 20% testing, and the accuracy is calculated per subject. On the other hand, the subject-independent approach allows to directly test all data points of a single subject with the model trained using the data of all other subjects, so a leave-one-subject-out approach is adopted. In this case, the training is done using the data of 10 subjects and testing using one subject. This process is repeated for all subjects such that each subject is used as a test-subject once.

III. Results

A. Subject-dependent SSVEP Recognition

Brain signals are weak and so variable that a classifier will not perform as well if trained and tested the next day or tested on a different subject [31]. Therefore, BCI studies are typically performed in a subject-dependent manner: Both training and testing are performed using the data of the same subject. In this study, we first explore the performance of the weighted and unweighted ensemble classifiers in comparison to the individual pre-processing and classifier combinations within subjects using the whole 5 sec stimulation time. We examined the training accuracy achieved using each pre-processing approach (No PCA or CAR, CAR only, PCA only, and PCA and CAR) for each of the two examined classifiers (SVM and Random Forest), and we present the results as bar graphs representing the mean accuracy computed across the 11 subjects and the standard deviation. Fig. 1 demonstrates that the CAR filter pre-processing and the Random Forest classifier

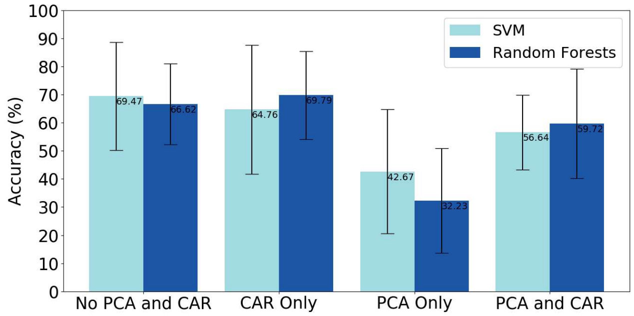

Fig. 1. Subject-dependent Mean Training Accuracy for 5 Seconds Stimulation Time for Each Pre-processing Method and Classifier (mean ± std).

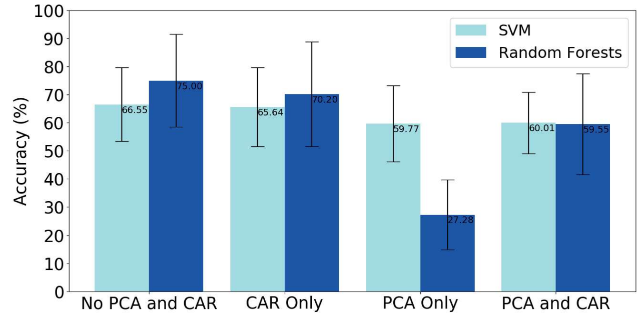

Fig. 3. Subject-dependent Mean Training Accuracy for 2.5 Seconds Stimulation Time for Each Pre-processing Method and Classifier (mean ± std).

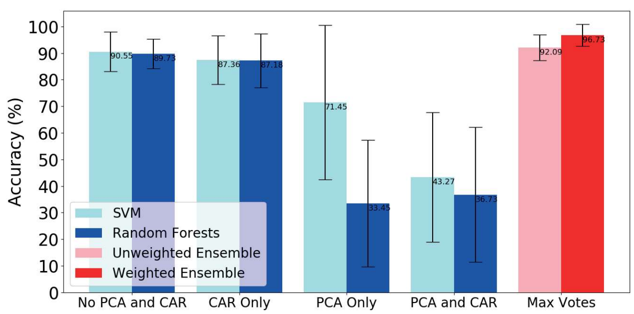

Fig. 2. Subject-dependent Mean Testing Accuracy for 5 Seconds Stimulation Time for Each Pre-processing Method and Classifier (mean ± std).

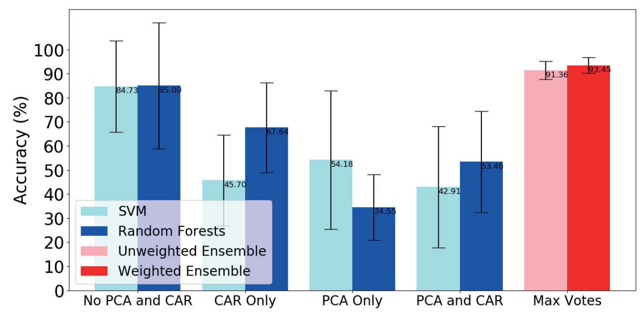

Fig. 4. Subject-dependent Mean Testing Accuracy for 2.5 Seconds Stimulation Time for Each Pre-processing Method and Classifier (mean ± std).

give the highest mean accuracy across all subjects of 69.79%. The figure also demonstrates variability across subjects as manifested by the increased standard deviation for all methods as the performance is highly affected by the subject.

We next examined the performance of the weighted and unweighted ensemble classifiers in comparison to the individual pre-processing and classification techniques when applied to the testing data. For the individual pre-processing and classification techniques combinations, the highest mean accuracy reached 90.55% which is achieved using the SVM classifier and no pre-processing as shown in Fig. 2. On the other hand, the unweighted ensemble classifier gives a mean accuracy of 92.09% and the weighted ensemble classifier gives a higher mean accuracy of 96.73%. These results demonstrate that the weighted ensemble classifier produces better predictions by giving a higher weight to the models with higher training accuracy while taking into account the models with lower training accuracy as well to compensate for overfitting.

The results of the 5 sec stimulation time encouraged exploring changes in the performance if we reduce the stimulation time. This is important as it could reduce the time needed to record data from the subjects and thus enhance the usability of SSVEP-based BCI systems. In this case, the eight models were re-trained using data that corresponds to windows of duration 2.5 sec. Fig. 3 shows the mean training accuracy for each model with the highest mean accuracy of 75% achieved without using any pre-processing (consistent with the results of the 5 sec stimulation time) and the Random Forest classifier.

Although the highest training accuracy in the case of 2.5 sec stimulation time (75%) is higher than that achieved for the 5 sec stimulation time (69.79%), such increase is not statistically significant. In fact, the testing accuracy for the 2.5 sec stimulation time is lower than its 5 sec stimulation time counterpart as demonstrated in Fig. 4. The individual model with the highest testing accuracy in this case is the Random Forest with no pre-processing achieving a mean accuracy of 85%. Employing the proposed ensemble classification technique had a significant impact on the testing classification accuracy. The unweighted ensemble classifier achieved a mean accuracy of 91.36%, while the weighted ensemble classifier achieved a mean accuracy of is 93.45%. These results are consistent with the conclusion from the 5 sec stimulation time. In addition, for the both the 5 sec and 2.5 sec stimulation times, the ensemble classifiers results have much lower variability across subjects compared to all other approaches examined.

B. Subject-independent SSVEP Recognition

We next examined the feasibility of using the SSVEP ensemble classifier universally across subjects. The main objective of subject-independent recognition is to decrease or completely eliminate the calibration time needed to record training data from each subject which is typically time-consuming and exhaustive. Similar to our analysis of the subject-dependent data, we first investigated the training and

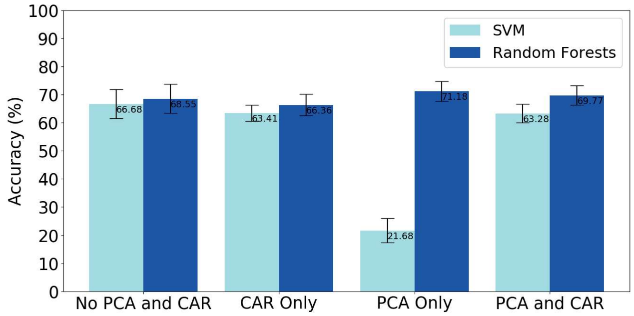

Fig. 5. Subject-independent Mean Training Accuracy for 5 Seconds Stimulation Time for Each Pre-processing Method and Classifier (mean ± std).

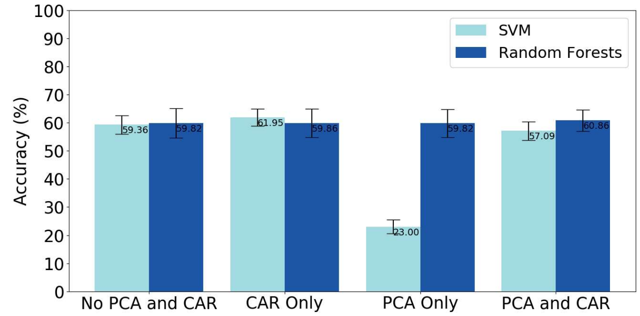

Fig. 7. Subject-independent Mean Training Accuracy for 2.5 Seconds Stimulation Time for Each Pre-processing Method and Classifier (mean ± std).

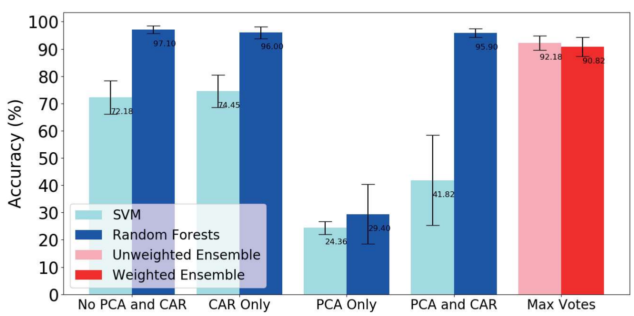

Fig. 6. Subject-independent Mean Testing Accuracy for 5 Seconds Stimulation Time for Each Pre-processing Method and Classifier (mean ± std).

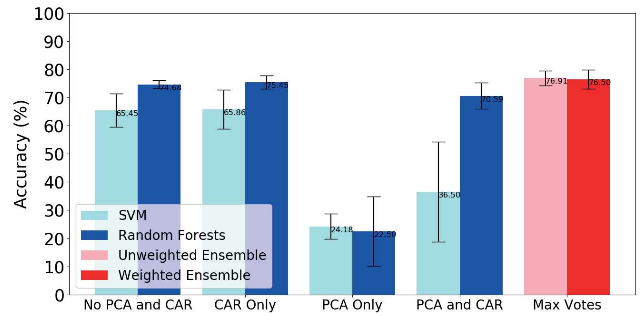

Fig. 8. Subject-independent Mean Testing Accuracy for 2.5 Seconds Stimulation Time for Each Pre-processing Method and Classifier (mean ± std).

testing performance when using the 5 sec stimulation time. Fig. 5 demonstrates that moving to a universal classifier across subjects results in an overall reduction in the achieved accuracy with the highest mean accuracy of 71.18% achieved using PCA pre-processing and Random Forest classifier.

Inconsistent with the subject-dependent results, the model with the highest training accuracy does not necessarily provide the highest testing accuracy as observed from Fig. 6, which increases the ambiguity when choosing the best model to proceed with in testing. In fact, the results demonstrated in Fig. 5 to Fig. 10 are not consistent, where in some cases the SVM classifier provides the highest accuracy, while in other cases the Random Forest classifier does so with different pre-processing. In this case, the ensemble classifier is utilized to provide stable testing results by taking all different pre-processing and classification techniques into account. The unweighted and weighted ensemble classifiers in this case resulted in a mean testing accuracy of 92.18% and 90.82%, respectively.

We finally examined the subject-independent results for the 2.5 sec stimulation time data. As shown in Fig. 7, the highest mean training accuracy is 61.95% achieved using the CAR filter and SVM. Similar to the results of the 5 sec stimulation time across subjects, the model with the highest training accuracy does not provide the highest testing accuracy as shown in Fig. 8. The unweighted ensemble classifier achieves a mean testing accuracy of 76.91%, while the weighted ensemble classifier achieves a mean testing accuracy of 76.5%. This confirms the utility of the proposed ensemble classifier for SSVEP-based BCIs.

IV. Conclusion

This study introduced an ensemble classification approach for SSVEP stimulus recognition. The approach is examined under different conditions including using the whole 5 sec stimulation time within and across subjects and using only 2.5 sec stimulation time within and across subjects. The ensemble classification approach focuses on increasing the predictive performance of the trained classifiers based on their training accuracy. There are two types of proposed ensemble classifiers: unweighted and weighted. The weighted ensemble classifier performed best in comparison to the individual classifiers as well as the unweighted ensemble classifier within subjects and increased the overall testing accuracy. Applying the weighted ensemble classifier to the 5 sec stimulation time data within subjects resulted in a mean accuracy of 96.73%, whereas for the 2.5 sec stimulation time data within subjects, a mean accuracy of 93.45% was achieved. On the other hand, the unweighted ensemble classifier performed best across subjects achieving a mean accuracy of 92.18% and 76.91% for the 5 sec and 2.5 sec stimulation times, respectively. These results demonstrate the utility of the proposed ensemble classifier in maximizing and stabilizing the overall accuracy by optimally combining the outcome of the different classifiers and pre-processing approaches examined.

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